

There's a formal name for the class of problems whose solutions can be verified in polynomial time: the *NP complexity class*. (Here, NP stands for *nondeterministic polynomial time*, for reasons that would require a long digression into theoretical computer science that would not be useful here.) NP is one of the two most fundamental complexity classes in computer science. The second is called *P*, for polynomial time. The P complexity class of problems contains all problems whose solutions can be found by an algorithm that runs in polynomial time. For P problems, we can *find* full solutions in polynomial time, while for NP problems, we can *verify* solutions in polynomial time, but it may take exponential time to find those solutions.

We know that sudoku is an NP problem—it is easy to verify a proposed sudoku solution in polynomial time. Is sudoku also a P problem? That is, is there an algorithm that can solve any sudoku puzzle in polynomial time? No one has ever found one, and no one appears to be close to finding one, but we don't feel certain that it's impossible.

The list of problems that we know are in NP is extremely long. Some versions of the traveling salesman problem are in NP. So is the optimal solution to the Rubik's cube, as well as important mathematical problems like integer linear programming. Just as with sudoku, we wonder whether these problems are also in P—can we find solutions for them in polynomial time? One way to phrase this question is, Does $P = NP$?

In 2000, the Clay Mathematics Institute published a list called the Millennium Prize Problems. It announced that any person who published a verified solution to one of the problems would receive a million dollars. The list was meant to be seven of the world's most important problems related to mathematics, and the question of whether $P = NP$ is one of them; no one has claimed its prize yet. Will one of the noble adventurers reading these words eventually break the Gordian knot and solve this most crucial of algorithmic problems? I sincerely hope so and wish each of you luck, strength, and joy on the journey.

If there is ever a solution, it will be a proof of one of the following two assertions: either that $P = NP$ or that $P \neq NP$. A proof that $P = NP$ could be relatively simple, since all that would be required is a polynomial-time algorithmic solution to an NP-complete problem. *NP-complete* problems